

LLM Fundamentals — Technical Notes

1. Neural Network — Artificial Neural Network (ANN)

Definition

Artificial Neural Network (ANN) is a computational model inspired by the structure of biological neural networks. It consists of interconnected computational units called **neurons/nodes**, organized into layers.

Simple-ah:

Human brain-la neurons irukkura concept-ai inspiration-ah eduthu, mathematical operations use pannu input-la irundhu output learn panna design pannina computational model dhaan Neural Network.

Basic Architecture

```
graph TD;
    A[Input Layer] --> B[Hidden Layer(s)];
    B --> C[Output Layer];
```

Example:

```
graph TD;
    A[Input:  
Image / Text / Numbers] --> B[Neural Network];
    B --> C[Output:  
Prediction / Classification];
```

Example

Input:

"Apple"

Model context-ai based on:

Apple → Fruit

nu classification/prediction panna mudiyum.

But important:

Neural Network "Apple = Fruit" nu manually store pannadhu illa. Training moolama parameters/weights learn pannum.

Important Technical Terms

Neuron / Node → Mathematical computation perform pannra unit.

Layer → Multiple neurons organized into a stage.

Weights → Input information-ku model kudukkura learned numerical importance.

Bias → Neuron output-ai adjust panna use pannra learned parameter.

Activation Function → Neural network-la non-linearity introduce panna use pannra mathematical function.

Examples:

- ReLU
- Sigmoid
- Tanh
- GELU

Parameters → Model training moolama learn pannra weights and biases.

2. Dataset — Training Data

Definition

Dataset is a collection of data used to train, validate, or evaluate a machine learning model.

LLM training-ku massive amounts of text data use pannuvanga.

Possible Data Sources

- Books
- Wikipedia
- Websites
- News articles
- Code repositories
- Public documents
- Other text datasets

Data formats:

- JSON
- JSONL
- TXT
- CSV
- Parquet
- HTML
- etc.

Important Technical Term

Corpus

NLP context-la large collection of text documents/data-ai **Corpus** nu solluvanga.

Example:

```
Documents
  ↓
Text Corpus
  ↓
Training Dataset
```

3. Data Cleaning / Data Preprocessing

Definition

Data Cleaning is the process of identifying and removing or correcting low-quality, duplicate, irrelevant, corrupted, or unwanted data before training.

Simple-ah:

Training-ku use panna pora raw data-va quality improve pannradhu dhaan Data Cleaning.

Common Cleaning Operations

```
Raw Data
  ↓
Remove Duplicates
  ↓
Remove Spam
  ↓
Remove Invalid Data
  ↓
Remove Unwanted HTML
```

↓
Quality Filtering
↓
Clean Dataset

Why?

Machine Learning-la:

Garbage In → Garbage Out

Low-quality training data model quality-ai affect pannalam.

Important Terms

Deduplication → Duplicate content/data remove pannradhu.

Data Filtering → Unwanted or low-quality data remove pannradhu.

Normalization → Data-ai consistent format-ku convert pannradhu.

Data Preprocessing → Training-ku suitable format-la raw data-ai prepare pannra overall process.

4. Tokenization

Definition

Tokenization is the process of converting text into smaller units called **tokens**, which can then be mapped to numerical token IDs for model processing.

Important:

Tokenization always means "words-ai numbers-a convert pannradhu" nu mattum solla koodadhu.

Tokens can represent:

- Words
- Subwords
- Characters
- Punctuation
- Special tokens

Modern LLMs usually use **subword/token-based tokenization**.

Example

Sentence:

"I love Python"

Possible tokens:

["I", "love", "Python"]

Then token IDs:

[40, 821, 14562]

Actual token IDs tokenizer/model-ku depend aagum.

Technical Pipeline

```
Raw Text
  ↓
Tokenizer
  ↓
Tokens
  ↓
Token IDs
  ↓
Neural Network
```

Important Terms

Token → Model process panna use pannra text unit.

Tokenizer → Text-ai tokens/token IDs-ah convert pannra component.

Token ID → Vocabulary-la oru token-ku assigned numerical identifier.

Vocabulary → Tokenizer recognize panna available tokens-oda collection.

5. Pre-training

Definition

Pre-training is the large-scale training process in which a neural network learns general patterns and representations from a massive dataset.

For many modern LLMs, a common objective is:

Next-token prediction

Earlier "Next Word Prediction" nu solluvom. Technical-ah **Next Token Prediction** nu solradhu more accurate.

Example

Input:

"Twinkle twinkle little"

Model prediction:

"star"

But training-la model just one time guess pannaadhu.

```
Input
↓
Prediction
↓
Compare with Target
↓
Calculate Loss
↓
Backpropagation
↓
Update Parameters
↓
Repeat
```

Indha process billions/trillions of tokens-ku repeat aagum.

6. Loss Function

Definition

Loss Function measures how different the model's prediction is from the expected target.

Simple-ah:

Model evlo wrong-ah predict pannuchu nu numerical value-la measure pannradhu dhaan Loss.

Example:

Target:
"star"

Model Prediction:
"moon"

↓

High Loss

Correct prediction probability increase aana:

Lower Loss

7. Backpropagation

Definition

Backpropagation is the algorithm used to calculate how each model parameter contributed to the prediction error.

Simple-ah:

Model wrong answer kudutha, andha error-ai backward direction-la calculate panni, endha parameters evlo change aaganum nu determine pannradhu dhaan Backpropagation.

Pipeline:

```
Prediction
↓
Loss Calculation
↓
Backpropagation
↓
Gradients
↓
Parameter Update
```

8. Gradient Descent / Optimizer

Gradient

Gradient tells the direction in which a model parameter should change to reduce the loss.

Optimizer

Optimizer uses gradients to update model parameters during training.

Common optimizers:

- SGD
- Adam
- AdamW

Basic idea:

```
Loss
  ↓
Gradient
  ↓
Optimizer
  ↓
Updated Parameters
```

9. Epoch, Batch and Training Step

Batch

Training data-la oru small group of samples/tokens process pannradhu.

Training Step / Iteration

One batch process panni model parameters update pannra oru training iteration.

Epoch

Entire training dataset-ai one complete pass process pannradhu.

Example:

```
Dataset
= 1,000,000 samples

Batch Size
```

= 1,000

1 Epoch

= Entire 1,000,000 samples processed

10. GPU / TPU

Large neural networks train panna huge computational power thevai.

GPU

Graphics Processing Unit

Parallel mathematical operations efficiently perform panna useful.

Examples:

- NVIDIA A100
- NVIDIA H100
- NVIDIA H200

TPU

Tensor Processing Unit

Google design panna machine-learning-focused accelerator.

Large-scale model training-la GPU/TPU clusters use pannuvanga.

11. Transformer Architecture

Modern LLMs-oda most important architecture:

Transformer

Transformer architecture was introduced in the research paper:

"Attention Is All You Need" (2017)

Transformer architecture-la major concepts:

- Self-Attention
- Multi-Head Attention
- Feed-Forward Network
- Positional Information

- Layer Normalization
- Residual Connections

Simplified LLM Flow

```
Text
↓
Tokenization
↓
Token IDs
↓
Embeddings
↓
Transformer Layers
↓
Output Logits
↓
Next Token Prediction
↓
Generated Text
```

12. Embedding

Definition

Embedding is a dense numerical representation of data such as tokens or text in a high-dimensional vector space.

Example:

```
Token
↓
Embedding Layer
↓
Vector

"king"
↓
[0.21, -0.43, 0.87, ...]
```

The actual vector dimensions depend on the model.

Important

Embedding has different contexts.

Token Embedding

Token → Vector

Used inside neural network.

Text Embedding

Text / Sentence / Document → Vector

Commonly used for:

- Semantic Search
- RAG
- Recommendation
- Similarity Search

13. NLP — Natural Language Processing

NLP = Natural Language Processing

Computer systems text and human language-ai process/understand/generate panna use pannra field.

NLP topics:

- Tokenization
- Text Classification
- Named Entity Recognition
- Sentiment Analysis
- Machine Translation
- Information Extraction
- Text Generation

Modern LLMs NLP-ai much more powerful-ah handle pannum.

14. Deep Learning

Deep Learning is a subset of Machine Learning that uses neural networks with multiple layers to learn complex patterns from data.

Hierarchy:

Artificial Intelligence



Machine Learning



Deep Learning



Neural Networks



Transformer



Modern LLMs

Beginner Roadmap

Step 1 — Python

First learn:

- Variables
- Data Types
- Lists
- Dictionaries
- Loops
- Conditions
- Functions
- Classes
- File Handling
- Exception Handling
- Modules
- Virtual Environments

Step 2 — Mathematics

Basic level-Ia:

- Linear Algebra
 - Vectors
 - Matrices
- Matrix Multiplication
- Probability
- Statistics
- Calculus basics
- Derivatives

- Gradients

LLM research level-ku pogumbothu math knowledge deeper-ah thevai.

Step 3 — NumPy / Pandas

Learn:

NumPy

→ Numerical computation and arrays.

Pandas

→ Structured data manipulation and analysis.

Step 4 — Machine Learning

Learn concepts:

- Dataset
- Features
- Labels
- Training
- Validation
- Testing
- Loss
- Overfitting
- Underfitting
- Model Evaluation
- Regression
- Classification

Library:

Scikit-learn

Step 5 — Deep Learning

Learn:

- Neural Networks
- Forward Propagation
- Loss Function
- Backpropagation

- Gradient Descent
- Optimizers
- Activation Functions
- CNN basics
- RNN basics

Frameworks:

- PyTorch
- TensorFlow

For modern LLM learning, **PyTorch** is especially useful.

Step 6 — NLP

Learn:

- Tokenization
- Embeddings
- Text Classification
- Language Models
- Attention
- Transformers

Libraries:

- Hugging Face Transformers
 - NLTK
 - spaCy
-

Step 7 — Transformers

This is where modern LLM architecture becomes clear.

Learn:

```
Token
↓
Embedding
↓
Positional Information
↓
Self-Attention
↓
Multi-Head Attention
↓
Feed-Forward Network
↓
```

Transformer Blocks
↓
Output

Important paper:

"Attention Is All You Need"

Complete Concept Map

```
graph TD;
  AI[Artificial Intelligence] --> ML[Machine Learning];
  ML --> DL[Deep Learning];
  DL --> NN[Neural Networks];
  NN --> T[Transformer];
  T --> LM[Language Model];
  LM --> LLM[LLM];
  LLM --> PT[Pre-training];
  LLM --> FT[Fine-Tuning];
  LLM --> GK[LLM + External Knowledge];
  PT --> GSC[General Capabilities];
  FT --> S[Specialized];
  GK --> RAG[RAG];
```

Pre-training Fine-Tuning ↓ ↓ General Specialized Capabilities Behavior

LLM + External Knowledge ↓ RAG

One-Line Technical Definitions

Neural Network → Learnable mathematical function composed of interconnected layers/parameters.

Dataset → Collection of data used for model training/evaluation.

Corpus → Collection of text documents used in NLP.

Data Preprocessing → Preparing raw data for model consumption.

Tokenization → Converting text into model-processable tokens.

Token ID → Numerical identifier assigned to a token.

Embedding → Dense numerical representation of tokens/text.

Parameter → Learnable numerical value in a model.

Loss → Numerical measurement of prediction error.

Backpropagation → Computing gradients of loss with respect to model parameters.

Gradient → Directional information used to reduce loss.

Optimizer → Algorithm that updates model parameters using gradients.

Pre-training → Large-scale initial training of a model on massive datasets.

Fine-Tuning → Additional training of an existing model for specific tasks/behavior.

RAG → Retrieving external information and providing it as context to an LLM before generation.

Transformer → Neural-network architecture based heavily on attention mechanisms, widely used in modern LLMs.

LLM → Large Language Model trained to model/generate language at large scale.

Inference → Using a trained model to generate predictions/outputs for new input.